SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

TradeLinker — Connecting People with Trusted Professionals

Project Name: TradeLinker
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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements of TradeLinker. This document describes the system's objectives, features, users, services, limitations, and technical requirements.

TradeLinker aims to provide a digital platform where individuals, hospitals, companies, and other organizations can find and connect with skilled professionals for different types of services.

1.2 Scope

TradeLinker is a proposed online service marketplace accessible through a website and mobile application. The platform will allow customers to search for service providers, view their profiles, compare services, and submit booking requests.

Service providers will be able to register, create professional profiles, list their services, manage booking requests, and communicate with customers.

The platform will initially focus on six major service categories:

1. Construction Services
2. Plumbing Services
3. Electrical Services
4. AC and HVAC Services
5. Solar Energy Services
6. Wooden Door Making and Carpentry

These categories represent the initial service focus of TradeLinker. Additional categories may be introduced in future versions.

1.3 Objectives

The main objectives of TradeLinker are:

- To connect customers with relevant skilled professionals.
- To simplify the process of finding and booking services.
- To provide professional profiles and service information.
- To help service providers reach more potential customers.
- To support communication between customers and professionals.
- To improve transparency through reviews and ratings.
- To provide an administrative system for managing users, services, and reported issues.
- To create opportunities for future expansion into additional service categories and locations.

1.4 Definitions and Abbreviations

- **SRS:** Software Requirements Specification.
- **UI:** User Interface.
- **UX:** User Experience.
- **Admin:** System administrator responsible for managing the platform.
- **Customer:** A person or organization looking for a service.
- **Service Provider:** A professional or business offering services.
- **MVP:** Minimum Viable Product.
- **Booking Request:** A request submitted by a customer for a service.
- **Verification:** A process used to check a provider's submitted information.

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2. Overall Description

2.1 Product Perspective

TradeLinker will operate as an independent online service marketplace. It will consist of a customer-facing interface, a service provider dashboard, and an administrative dashboard.

The system will store user accounts, professional profiles, service listings, booking requests, reviews, and other relevant information in a central database.

2.2 Product Functions

The major functions of TradeLinker will include:

- User registration and login.
- Customer and service provider profile management.
- Service category browsing.
- Search and filtering of professionals.
- Service listing management.
- Booking request submission and management.
- Customer-provider communication.
- Chat functionality for communication between users.
- Call functionality, depending on the implemented communication features.
- Story functionality for sharing service updates and professional work, if included in the initial release.
- Reviews and ratings.
- Notifications regarding bookings and other important activities.
- Administrative management.
- Reporting and complaint handling.

TradeLinker will combine service discovery, booking management, and communication features in one platform. Its communication interface will allow users to interact with relevant customers or service providers, subject to the platform's privacy and access rules.

2.3 User Classes

1. Customers

Customers may include individuals, hospitals, companies, and other organizations. They can search for professionals, review service details, submit booking requests, and rate completed services.

2. Service Providers

Service providers may include skilled workers, technicians, contractors, carpenters, and registered service businesses. They can manage profiles, publish services, receive booking requests, and update request statuses.

3. Administrator

The administrator manages user accounts, provider verification, service categories, reported content, and general platform operations.

4. Organization Accounts

Organizations may use the platform to find professionals for maintenance, construction, electrical work, plumbing, and other operational needs. Organization-specific features may be introduced in later versions.

2.4 Operating Environment

TradeLinker is planned to support:

- Modern web browsers, including Google Chrome, Microsoft Edge, and Mozilla Firefox.
- Desktop and laptop computers.
- Android mobile devices.
- Responsive layouts for mobile, tablet, and desktop screens.
- An internet connection for accessing online features.

2.5 Constraints and Assumptions

- Users must have internet access to use the platform.
- Users must provide valid registration information.
- Service availability depends on registered providers in the selected location.
- Provider verification depends on the information and documents submitted.
- SMS, email, payment, mapping, and other external integrations may require third-party services.

- The initial release will prioritize the six specified service categories.
- The first release may focus on a selected geographical area before expanding to other locations.

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3. Functional Requirements

3.1 User Registration and Authentication

FR-01: The system shall allow customers to create accounts using the required registration details.

FR-02: The system shall allow service providers to register separately from customers.

FR-03: The system shall allow registered users to log in and log out securely.

FR-04: The system shall validate required fields and prevent duplicate accounts where applicable.

FR-05: The system shall provide a password reset process.

FR-06: The system shall allow users to update their account information.

3.2 Customer Profile Management

FR-07: The system shall allow customers to create and edit their profiles.

FR-08: The system shall allow customers to provide their name, contact information, and service location.

FR-09: The system shall allow customers to view their booking history and current booking requests.

FR-10: The system shall allow customers to cancel eligible booking requests according to the platform's cancellation rules.

3.3 Service Provider Management

FR-11: The system shall allow providers to create professional profiles.

FR-12: The system shall allow providers to add their skills, experience, service areas, descriptions, and contact details.

FR-13: The system shall allow providers to upload relevant profile images and supporting documents where required.

FR-14: The system shall allow providers to create, update, and remove their service listings.

FR-15: The system shall allow providers to specify their service charges or indicate that prices are available upon request.

FR-16: The system shall allow providers to manage incoming booking requests.

FR-17: The system shall allow providers to update booking statuses according to permitted workflow rules.

FR-18: The system shall support an administrator-controlled provider verification process.

3.4 Service Categories

FR-19: The system shall display the six initial service categories.

FR-20: The system shall allow customers to browse services within each category.

FR-21: The system shall display relevant service information, including descriptions, provider details, and available pricing information.

FR-22: The administrator shall be able to add, update, activate, or deactivate service categories.

Initial Service Categories

1. Construction Services: Construction workers, contractors, civil engineers, and related professionals.

2. Plumbing Services: Pipe installation, water supply systems, drainage, and plumbing repairs.

3. Electrical Services: Electrical installation, maintenance, wiring, and troubleshooting.

4. AC and HVAC Services: Air conditioning installation, maintenance, repair, and ventilation services.

5. Solar Energy Services: Solar panel installation, maintenance, and related system services.

6. Wooden Door Making and Carpentry: Wooden door manufacturing, fitting, repairs, and carpentry work.

3.5 Search and Filtering

FR-23: The system shall allow customers to search for professionals and services using keywords.

FR-24: The system shall allow customers to filter results by category and location.

FR-25: The system may support additional filters, such as price range, experience, availability, and rating.

FR-26: The system shall display relevant results based on the customer's search criteria.

FR-27: The system shall display an appropriate message when no matching results are found.

3.6 Booking Management

FR-28: The system shall allow customers to submit booking requests to service providers.

FR-29: The booking request shall include the selected service, problem description, service location, and preferred date or time when applicable.

FR-30: The system shall assign each booking a unique identifier.

FR-31: The system shall allow providers to accept or reject booking requests.

FR-32: The system shall allow customers to view booking status updates.

FR-33: The system shall support the following booking statuses:

- Pending
- Accepted
- Rejected
- In Progress
- Completed
- Cancelled

FR-34: The system shall maintain a record of booking status changes.

3.7 Communication and Notifications

FR-35: The system shall provide a way for customers and providers to communicate about service requests.

FR-36: The system shall notify users of important booking updates.

FR-37: Notifications may be delivered through in-app messages, email, or push notifications, depending on the implemented integrations.

FR-38: The system shall allow users to view relevant booking notifications.

3.8 Reviews and Ratings

FR-39: The system shall allow eligible customers to submit ratings and reviews after a completed service.

FR-40: The system shall display provider ratings and customer reviews on relevant profiles.

FR-41: The system shall allow administrators to review reported or inappropriate content.

FR-42: The system shall restrict reviews to eligible users according to platform rules.

3.9 Administrator Dashboard

FR-43: The system shall provide a secure administrative dashboard.

FR-44: The administrator shall be able to view and manage user accounts.

FR-45: The administrator shall be able to review provider verification submissions.

FR-46: The administrator shall be able to manage service categories and listings.

FR-47: The administrator shall be able to investigate complaints and reported content.

FR-48: The administrator shall be able to monitor booking activity and basic platform statistics.

FR-49: The system shall restrict administrative functions to authorized administrators.

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4. Non-Functional Requirements

4.1 Performance

NFR-01: The system should load ordinary pages within approximately three seconds under normal operating conditions.

NFR-02: The system should process common search and booking operations efficiently.

NFR-03: The system should support increasing numbers of users and service listings as the platform grows.

4.2 Security

NFR-04: The system shall use secure authentication and appropriate password protection.

NFR-05: The system shall use HTTPS for data transmission in production.

NFR-06: The system shall restrict access to private information based on user roles.

NFR-07: The system shall validate user inputs and protect against common web application vulnerabilities.

NFR-08: The system shall protect personal information and only collect information necessary for its functions.

NFR-09: The system shall provide appropriate controls for account access and administrative operations.

4.3 Usability

NFR-10: The user interface shall be simple, clear, and easy to navigate.

NFR-11: The website shall use responsive layouts for desktop, tablet, and mobile screens.

NFR-12: The system shall display understandable error messages and confirmation messages.

NFR-13: Users should be able to find relevant services with minimal unnecessary steps.

4.4 Reliability and Availability

NFR-14: The system should operate reliably under expected usage conditions.

NFR-15: The system should maintain data consistency during registration, booking, and profile updates.

NFR-16: The system should support regular database backups and recovery procedures.

4.5 Maintainability

NFR-17: The system should use a modular structure that allows developers to maintain and extend its features.

NFR-18: The system should follow consistent coding standards and include appropriate technical documentation.

NFR-19: New service categories should be added without requiring a complete redesign.

4.6 Compatibility

NFR-20: The website shall support current versions of commonly used browsers.

NFR-21: The user interface shall adapt to different screen sizes.

NFR-22: The backend shall support integration with future mobile applications and external services through documented interfaces where appropriate.

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5. External Interface Requirements

5.1 User Interface

The platform shall provide the following primary interfaces:

- Home page
- Registration and login pages
- Service categories page
- Search results page
- Service provider profile page
- Customer dashboard
- Service provider dashboard
- Booking management page
- Notifications and communication interface
- Reviews and ratings interface
- Administrator dashboard
- Contact and support page

5.2 Software Interfaces

The system may integrate with:

- A relational database management system.
- An email delivery service.
- An SMS or notification service.
- A mapping service for location-based search.
- A payment gateway if online payments are introduced.
- Mobile push notification services in future versions.

5.3 Communication Interfaces

The system shall use secure HTTP communication through HTTPS in production. Any external integration shall follow the relevant provider's security and technical requirements.

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6. Data Requirements

The system shall maintain the following principal data entities.

6.1 User

- User ID
- Full name
- Email address
- Phone number
- Password hash
- User role
- Account status
- Registration date

6.2 Service Provider Profile

- Provider ID
- User ID
- Business or professional name
- Skills and experience
- Service categories
- Service locations
- Profile description
- Verification status

6.3 Service Listing

- Service ID
- Provider ID
- Category ID
- Service title
- Description
- Pricing information

- Service availability
- Listing status

6.4 Booking

- Booking ID
- Customer ID
- Provider ID
- Service ID
- Problem description
- Service location
- Preferred schedule
- Booking status
- Creation date
- Last updated date

6.5 Review

- Review ID
- Booking ID
- Customer ID
- Provider ID
- Rating
- Review text
- Submission date

6.6 Notification

- Notification ID
- Recipient user ID
- Notification type
- Message

- Read status
- Creation date

The system shall apply appropriate validation, access controls, and data relationships to maintain data integrity.

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7. Business Rules

BR-01: Users must register and authenticate before accessing account-specific functions.

BR-02: Customers may browse publicly available service listings without registering, unless a feature requires an account.

BR-03: Service providers must complete the required profile information before publishing services.

BR-04: A booking request must be associated with a valid customer, provider, and service.

BR-05: Only authorized users may change the status of a booking.

BR-06: Customers may submit reviews only when the relevant booking meets the platform's review eligibility rules.

BR-07: Provider verification does not guarantee the quality of every service; the platform shall communicate its verification scope clearly.

BR-08: The administrator may suspend accounts or remove listings that violate platform policies.

BR-09: The initial service focus shall remain on construction, plumbing, electrical, AC and HVAC, solar energy, and wooden door making and carpentry.

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8. Use Cases

UC-01: Customer Searches for a Service

Actor: Customer

Precondition: The platform is accessible.

Main Flow:

1. The customer opens the TradeLinker website or application.

2. The customer selects a service category or enters a search keyword.
3. The customer selects a location or applies filters.
4. The system displays matching service providers.
5. The customer views a provider's profile and service details.

Postcondition: The customer can identify a suitable provider or continue searching.

UC-02: Customer Books a Service

Actor: Customer

Precondition: The customer is logged in and the selected service is available for booking.

Main Flow:

1. The customer opens a service listing.
2. The customer selects the booking option.
3. The customer provides the required service details.
4. The customer submits the booking request.
5. The system creates a booking record.
6. The provider receives a notification.
7. The customer can view the booking status.

Postcondition: A booking request is recorded in the system.

UC-03: Provider Manages a Booking

Actor: Service Provider

Precondition: The provider is logged in and has received a booking request.

Main Flow:

1. The provider opens the dashboard.
2. The provider views pending booking requests.
3. The provider reviews the customer's service requirements.
4. The provider accepts or rejects the request.
5. The system updates the booking status.

6. The customer receives a status notification.

Postcondition: The booking request has an updated status.

UC-04: Administrator Verifies a Provider

Actor: Administrator

Precondition: A provider has submitted the required verification information.

Main Flow:

1. The administrator opens the verification section.
2. The administrator reviews the submitted information.
3. The administrator approves the provider, rejects the submission, or requests additional information.
4. The system records the decision.
5. The provider is notified of the outcome.

Postcondition: The provider's verification status is updated.

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9. Technology Stack

The following technology stack is proposed and may be adjusted according to project resources and development requirements.

Frontend

- HTML5
- CSS3
- JavaScript

Database and Backend Services

- Firebase or Supabase, depending on the selected architecture and project requirements.

Development and Deployment Tools

- Visual Studio Code
- Git and GitHub
- Vercel

The final technology stack shall be selected based on the team's skills, project budget, maintainability, scalability, and deployment requirements.

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10. Future Enhancements

The following features may be considered for future versions:

- Online payments and secure payment processing.
- Premium provider subscriptions.
- Featured service listings and promoted profiles.
- Business and organization accounts.
- Booking commissions where legally and commercially appropriate.
- Live location and map-based service discovery.
- Advanced provider availability scheduling.
- In-app chat and file sharing.
- AI-assisted service recommendations.
- Additional service categories.
- Expansion into more cities and international markets.
- Dedicated Android and iOS applications.
- Advanced voice and video calling features.
- Stories or temporary updates for showcasing completed work and services.

These features are planned possibilities and are not necessarily included in the initial release.

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11. Acceptance Criteria

The initial version of TradeLinker shall be considered acceptable when:

1. Customers and service providers can register and log in.
2. Providers can create profiles and publish service listings.
3. The six initial service categories are available.

4. Customers can search and filter service listings.
5. Customers can submit booking requests.
6. Providers can accept or reject booking requests.
7. Booking statuses are updated and displayed correctly.
8. Eligible customers can submit reviews and ratings.
9. Administrators can manage users, listings, and provider verification.
10. Access controls protect restricted information and administrative functions.
11. The main interfaces work on supported desktop and mobile browsers.
12. The main workflows pass the agreed functional and security tests.

Communication features such as chat, calls, and stories shall be included in the acceptance criteria if they are selected for the initial release.

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12. Conclusion

TradeLinker is intended to provide a centralized digital marketplace that connects customers and organizations with skilled professionals. The initial version will focus on six service categories: construction, plumbing, electrical, AC and HVAC, solar energy, and wooden door making and carpentry.

The system will provide user registration, professional profiles, service discovery, booking management, notifications, reviews, and administrative controls. Its modular design will support future improvements, additional service categories, premium business features, and expansion into new markets.

This SRS establishes the initial requirements and provides a foundation for system design, database development, implementation, and testing.

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